Multi-Currency Travel Budget Analysis

Complete worked solution

Use user-supplied units of local currency per one unit of foreign currency. For each expense, convert foreign amount by multiplying by its rate, then add a conversion fee percentage. Sum local totals, compare with the budget, and find the largest converted category. Rates are examples, not live exchange rates.

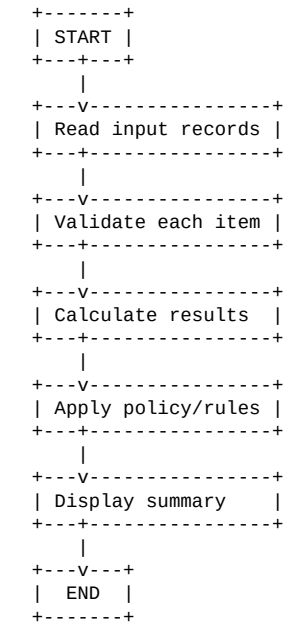
1. Computational Thinking and Algorithm

Inputs: Supplied records and policy values. **Processing:** Validate data, calculate charges or scores, apply ordered rules, and compare outcomes. **Outputs:** Itemized results, classifications, rankings, or recommendations.

Pseudocode

```
START
  Read budget, fee rate, and expense tuples
  FOR each expense
    Validate amount and exchange rate
    Convert to local currency and add fee
    Store converted category total
  Sum total and compare with budget
  Scan categories to find largest expense
  Display converted expense list, budget status, largest category
END
```

2. Flowchart



3. Python Program

The program uses basic variables, numeric and string data, tuples, lists, loops, conditionals, arithmetic, and formatted output. Policy values are constants near the top of the program.

```
# Each expense: (category, foreign_amount, local_per_foreign_unit)
expenses = [
    ("Flight", 420, 305.0),
    ("Hotel", 180, 305.0),
    ("Meals", 90000, 1.0),  # already in local currency
    ("Activities", 150, 305.0),
]
budget_local = 250000
```

```
conversion_fee_percent = 2.0
converted = []
total_spending = 0
for category, amount, exchange_rate in expenses:
    if amount < 0 or exchange_rate <= 0:
        print("Invalid expense:", category)
        continue
    base_local = amount * exchange_rate
    fee = base_local * conversion_fee_percent / 100
    final_local = base_local + fee
    converted.append((category, final_local))
    total_spending = total_spending + final_local
    print(category, "foreign amount:", amount, "rate:", exchange_rate,
          "fee:", round(fee, 2), "local total:", round(final_local, 2))
if total_spending <= budget_local:
    print("Within budget by:", round(budget_local - total_spending, 2))
else:
    print("Over budget by:", round(total_spending - budget_local, 2))
if len(converted) > 0:
    largest = converted[0]
    for expense in converted:
        if expense[1] > largest[1]:
            largest = expense
    print("Largest expense category:", largest[0],
          round(largest[1], 2))
```

4. Sample run guidance

Run the program as written to see a sample result. Replace the sample lists and constants with the values provided in your assignment. Keep the rule order and units consistent; for example, percentages should be entered as percentages and converted to fractions only in the calculation.